

Experiment No: 2

Exploratory Data Analysis (EDA) – Data Import and Export

Aim

To import the KaggleV2-May-2016.csv dataset using Pandas, inspect and analyze the dataset, perform basic exploratory data analysis (EDA), and export the processed data into different file formats.

Algorithm

1. Import the required Python libraries.
2. Load the healthcare dataset (`KaggleV2-May-2016.csv`) using `pd.read_csv()`.
3. Display the first few records using `head()`.
4. Display the dataset information using `info()`.
5. Display statistical summary using `describe()`.
6. Display the column names.
7. Check the shape of the dataset.
8. Check for missing values.
9. Check for duplicate records.
10. Display unique values of each column.
11. Visualize the relationship between selected variables using a scatter plot.
12. Separate the feature variables (`X`) and target variable (`y`).
13. Export the processed dataset into CSV, Excel, and JSON formats.

```
> # Step 1: Import Libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

Step 2: Upload Dataset in Google Colab

```
from google.colab import files
uploaded = files.upload()
```

Choose files KaggleV2-May-2016.csv

KaggleV2-May-2016.csv(text/csv) - 10739535 bytes, last modified: 23/07/2026 - 100% done
Saving KaggleV2-May-2016.csv to KaggleV2-May-2016 (1).csv

```
print(df.head())
```

	PatientId	AppointmentID	Gender	ScheduledDay	\
0	2.987250e+13	5642903	F	2016-04-29T18:38:08Z	
1	5.589978e+14	5642503	M	2016-04-29T16:08:27Z	
2	4.262962e+12	5642549	F	2016-04-29T16:19:04Z	
3	8.679512e+11	5642828	F	2016-04-29T17:29:31Z	
4	8.841186e+12	5642494	F	2016-04-29T16:07:23Z	

	AppointmentDay	Age	Neighbourhood	Scholarship	Hipertension	\
0	2016-04-29T00:00:00Z	62	JARDIM DA PENHA	0	1	
1	2016-04-29T00:00:00Z	56	JARDIM DA PENHA	0	0	
2	2016-04-29T00:00:00Z	62	MATA DA PRAIA	0	0	
3	2016-04-29T00:00:00Z	8	PONTAL DE CAMBURI	0	0	
4	2016-04-29T00:00:00Z	56	JARDIM DA PENHA	0	1	

	Diabetes	Alcoholism	Handcap	SMS_received	No-show
0	0	0	0.0	0.0	No
1	0	0	0.0	0.0	No
2	0	0	0.0	0.0	No
3	0	0	0.0	0.0	No
4	1	0	0.0	0.0	No

```
import os
```

```
print(os.listdir())
```

```
['.config', 'KaggleV2-May-2016.csv', 'sample_data']
```

```
print("\nDataset Information")
df.info()
```

```
Dataset Information
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 132084 entries, 0 to 132083
Data columns (total 14 columns):
#   Column                Non-Null Count  Dtype
---  ---
0   PatientId             132084 non-null float64
1   AppointmentID         132084 non-null int64
2   Gender                132084 non-null object
3   ScheduledDay          132084 non-null object
4   AppointmentDay        132084 non-null object
5   Age                  132084 non-null int64
6   Neighbourhood         132084 non-null object
7   Scholarship           132084 non-null int64
8   Hipertension          132084 non-null int64
9   Diabetes              132084 non-null int64
10  Alcoholism            132084 non-null object
11  Handcap                132083 non-null float64
12  SMS_received          132083 non-null float64
13  No-show               132083 non-null object
dtypes: float64(3), int64(5), object(6)
memory usage: 14.1+ MB
```

```
print("\nStatistical Summary")
print(df.describe())
```

```
Statistical Summary
```

	PatientId	AppointmentID	Age	Scholarship	\
count	1.320840e+05	1.320840e+05	132084.000000	132084.000000	
mean	1.472340e+14	5.672981e+06	36.963569	0.098263	
std	2.558273e+14	7.003391e+04	23.029998	0.297671	
min	3.921784e+04	5.030230e+06	-1.000000	0.000000	
25%	4.162841e+12	5.639690e+06	18.000000	0.000000	
50%	3.155544e+13	5.678315e+06	37.000000	0.000000	
75%	9.434637e+13	5.720900e+06	55.000000	0.000000	
max	9.999816e+14	5.790484e+06	115.000000	1.000000	

	Hipertension	Diabetes	Handcap	SMS_received
count	132084.000000	132084.000000	132083.000000	132083.000000
mean	0.196405	0.071008	0.022281	0.317936
std	0.397280	0.256839	0.161604	0.465676
min	0.000000	0.000000	0.000000	0.000000
25%	0.000000	0.000000	0.000000	0.000000
50%	0.000000	0.000000	0.000000	0.000000
75%	0.000000	0.000000	0.000000	1.000000
max	1.000000	1.000000	4.000000	1.000000

```
print("\nColumns in Dataset")
print(df.columns.tolist())
```

```
Columns in Dataset
['PatientId', 'AppointmentID', 'Gender', 'ScheduledDay', 'Appointment
```

```
print("\nShape of Dataset")
print(df.shape)

print("\nRows :", df.shape[0])
print("Columns :", df.shape[1])
```

Shape of Dataset
(132084, 14)

Rows : 132084
Columns : 14

```
print("\nRandom Sample")
print(df.sample(5, random_state=42))
```

```
Random Sample
PatientId AppointmentID Gender ScheduledDay \
27185 4.821340e+12 5312204 F 2016-02-01T14:11:49Z
9679 7.652494e+14 5663412 F 2016-05-05T09:59:31Z
85880 9.341366e+13 5679791 F 2016-05-10T10:30:07Z
42838 3.256279e+13 5618944 M 2016-04-26T07:33:24Z
48930 5.113857e+13 5680977 F 2016-05-10T13:21:40Z

AppointmentDay Age Neighbourhood Scholarship Hipertension \
27185 2016-05-05T00:00:00Z 41 RESISTÊNCIA 0 0
9679 2016-05-06T00:00:00Z 2 FORTE SÃO JOÃO 0 0
85880 2016-05-13T00:00:00Z 55 JARDIM CÂMBURI 0 1
42838 2016-05-05T00:00:00Z 40 SANTO ANDRÉ 0 0
48930 2016-05-10T00:00:00Z 36 SÃO CRISTÓVÃO 0 0

Diabetes Alcoholism Handcap SMS_received No-show
27185 0 0 0.0 1.0 No
9679 0 0 0.0 0.0 No
85880 1 0 0.0 0.0 No
42838 0 0 0.0 1.0 Yes
48930 0 0 0.0 0.0 No
```

```
missing = pd.DataFrame({
    "Missing Values": df.isnull().sum(),
    "Percentage": (df.isnull().sum()/len(df))*100
})

print("\nMissing Values")
print(missing)
```

Missing Values	Missing Values	Percentage
PatientId	0	0.000000
AppointmentID	0	0.000000
Gender	0	0.000000
ScheduledDay	0	0.000000
AppointmentDay	0	0.000000
Age	0	0.000000
Neighbourhood	0	0.000000
Scholarship	0	0.000000
Hipertension	0	0.000000
Diabetes	0	0.000000
Alcoholism	0	0.000000
Handcap	1	0.000757
SMS_received	1	0.000757
No-show	1	0.000757

```
duplicates = df.duplicated().sum()

print("\nDuplicate Rows :", duplicates)
```

Duplicate Rows : 21556

```
print("\nUnique Values in Each Column\n")

for column in df.columns:
    print(column, ":", df[column].nunique())
```

Unique Values in Each Column

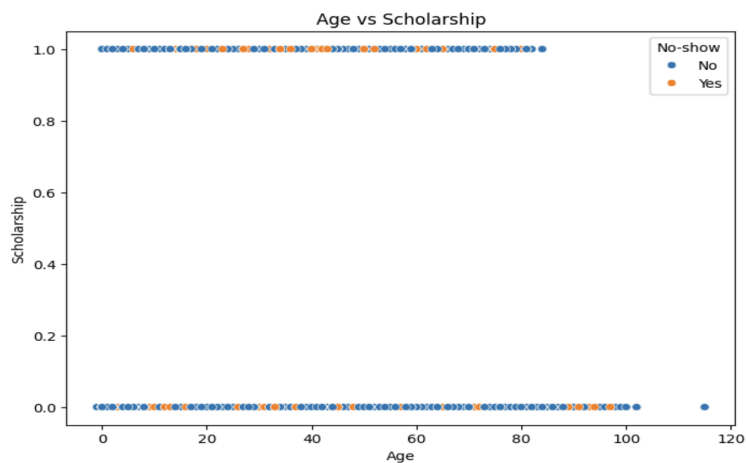
```
PatientId : 62299
AppointmentID : 110527
Gender : 2
ScheduledDay : 103550
AppointmentDay : 28
Age : 104
Neighbourhood : 82
Scholarship : 2
Hipertension : 2
Diabetes : 2
Alcoholism : 5
Handcap : 5
SMS_received : 2
No-show : 2
```

```
plt.figure(figsize=(8,6))

sns.scatterplot(
    data=df,
    x="Age",
    y="Scholarship",
    hue="No-show"
)

plt.title("Age vs Scholarship")
plt.xlabel("Age")
plt.ylabel("Scholarship")

plt.show()
```



```

numeric_df = df.select_dtypes(include=np.number)

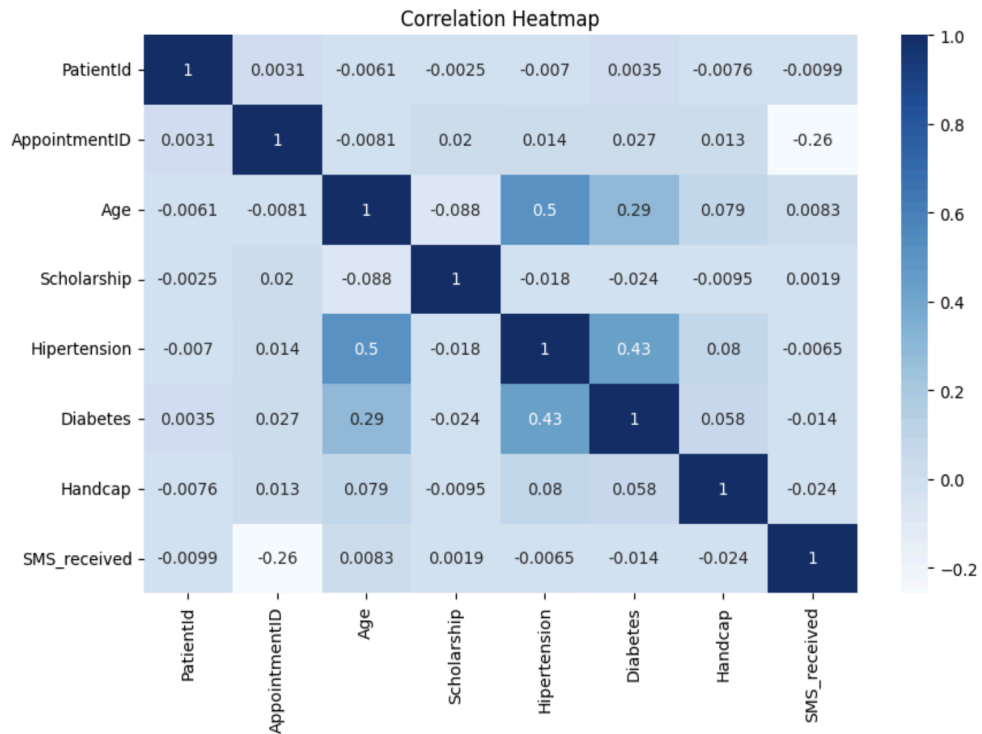
plt.figure(figsize=(10,6))

sns.heatmap(
    numeric_df.corr(),
    annot=True,
    cmap="Blues"
)

plt.title("Correlation Heatmap")

plt.show()

```



```

X = df.drop("No-show", axis=1)

y = df["No-show"]

print("\nFeature Data Types\n")
print(X.dtypes)

print("\nTarget Variable")
print(y.head())

```

Feature Data Types

```

PatientId          float64
AppointmentID      int64
Gender             object
ScheduledDay       object
AppointmentDay     object
Age               int64
Neighbourhood      object
Scholarship        int64
Hipertension       int64
Diabetes           int64
Alcoholism         object
Handcap           float64
SMS_received       float64
dtype: object

```

Target Variable

```

0      No
1      No
2      No
3      No
4      No
Name: No-show, dtype: object

```

```

df.to_csv("Processed_KaggleV2.csv", index=False)

df.to_excel("Processed_KaggleV2.xlsx", index=False)

df.to_json("Processed_KaggleV2.json")

print("\nDataset Exported Successfully")

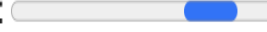
```

Dataset Exported Successfully


```
files.download("Processed_KaggleV2.csv")  
files.download("Processed_KaggleV2.xlsx")  
files.download("Processed_KaggleV2.json")
```

Downloading "Processed_KaggleV2.csv": 

Downloading "Processed_KaggleV2.xlsx": 

Downloading "Processed_KaggleV2.json": 

RESULT:

1. Thus, missing values, duplicate records, and outliers in the KaggleV2-May-2016.csv dataset were successfully identified and handled. Data transformation techniques such as normalization, label encoding, and one-hot encoding were performed successfully, preparing the dataset for further exploratory data analysis and machine learning applications.